

Multi-Perception Crowd: Learning to combine entity and implicit perception for diverse crowd simulation

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Abstract— We present a reinforcement learning framework for crowd simulation that balances collision avoidance with navigation preferences guided by soft environmental constraints. Our approach integrates two complementary perception components: *entity perception*, which handles hard constraints imposed by physical obstacles, and *implicit environmental perception*, encoded as suitability maps that guide movement preferences due to soft constraints. To ensure robust generalization across complex scenes, we employ a modular, two-phase training strategy utilizing curriculum-based environmental templates. The proposed framework functions as an intuitive crowd authoring tool: artists can influence crowd behavior in real time by interactively editing suitability maps or dynamically adjusting perception weights, either globally or per-agent. This enables adaptive navigation behaviors, such as dynamic trajectory prioritization based on the environment, to emerge from local interactions. We validate our approach using both quantitative metrics and qualitative analysis, including a user study confirming that the resulting behaviors align with pedestrian traffic patterns observed in real-world settings. We further present an example of emergent social behavior through the formation and gradual evolution of desire paths. This work contributes to the field of crowd simulation by offering a robust, learning-based framework that supports heterogeneous navigation modeling and intuitive authoring for interactive applications.

Index Terms—Crowd Simulation, Reinforcement Learning, Crowd Authoring, Animations

I. INTRODUCTION

Crowd simulation is a fundamental component of virtual environments, essential for applications ranging from urban planning [1] and virtual training [2] to entertainment [3]. While minimizing collisions and reaching destinations are the primary directives of agent navigation, generating diverse and context-aware behaviors requires modeling how individuals interact with the semantics of their environment. In complex scenarios, pedestrians do not merely navigate around physical obstacles; they adhere to implicit environmental rules, such as preferring shaded paths on hot days, avoiding puddles on a rainy day, or following crosswalks unless urgency dictates otherwise. These nuanced behaviors stem from a continuous trade-off between physical safety (hard constraints) and environmental preferences (soft constraints).

Traditional crowd simulation approaches typically rely on a decoupled pipeline: global path-finding on navigation meshes

for guidance, and local steering algorithms (rule-based [4]–[6] or force-based [7]–[10]) for collision avoidance. While effective for basic navigation, this paradigm treats the environment as a binary partition of traversable and non-traversable space. This rigid dichotomy inherently limits the simulation of soft constraints, as it fails to capture continuous preference gradients where traversal is physically possible yet contextually discouraged. Moreover, these deterministic algorithms often yield robotic or overly uniform crowd behaviors. To address the limitations of handcrafted rules, recent data-driven [11]–[13] and reinforcement learning (RL) methods [14]–[17] have been widely adopted. RL is particularly advantageous as it enables agents to learn complex, non-linear policies directly from interaction, optimizing long-term cumulative returns rather than just immediate greedy reactions. This capability allows for significantly improved naturalness and heterogeneity in local navigation dynamics. However, despite these advancements in motion naturalness, most existing RL-based approaches lack the mechanism to integrate implicit environmental preferences, effectively treating environmental semantics as irrelevant. This rigidity restricts agents from dynamically adapting to soft constraints, leaving them blind to the contextual nuances of the scene.

This technical inflexibility creates a disconnect between the agent’s automatic navigation and the designer’s intent. Consequently, creating context-aware crowds remains a laborious task for designers. While various authoring tools have been proposed to control macroscopic flows or densities [18], they often lack the granularity to model subtle environmental interactions. For instance, recent methods like MPACT [19] allow artists to mimic crowd steering from video data, but they focus exclusively on agent-agent interactions, effectively ignoring the environment. As a result, replicating context-dependent behaviors, such as pedestrians generally adhering to a path but temporarily stepping onto grass to bypass congestion, currently forces artists into a manual, laborious process of setting explicit waypoints or invisible barriers. To overcome this authoring bottleneck, implicit environmental rules must be integrated directly into the agent’s perception system, enabling dynamic simulations governed by soft constraints that respond naturally to artistic inputs without requiring manual micro-management.

In this paper, we present a novel Multi-Perception Reinforcement Learning framework that explicitly decouples the learning of physical constraints from environmental preferences. Our approach introduces two distinct perception modules (Figure 1): **Entity Perception**, which utilizes raycasting to detect physical colliders (agents and obstacles), and **Im-**

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Implicit Perception, which processes local "suitability maps" to encode semantic preferences. By representing implicit environmental rules as grayscale suitability maps, where intensity correlates with movement preference, we transform complex semantic constraints into a unified input format that is both computationally efficient and artistically intuitive.

To address the issue of feature interference during learning, we propose a modular two-phase training strategy. In the first phase, agents learn entity avoidance and implicit rule-following independently, preventing one objective from suppressing the other. In the second phase, a Perception Fusion Network learns to dynamically weight these inputs based on the agent's state. Furthermore, to ensure that agents can adaptively interpret topological structures (e.g., intersections, curved paths) from pixel data, we employ a curriculum-based env-templates that progressively increases environmental complexity. We validate the effectiveness of the entity and implicit perception control mechanisms through quantitative experiments, confirming the efficacy of the proposed training strategy via ablation studies, and demonstrate the model's robust generalization to unseen environments.

Moreover, this architecture serves as a robust authoring tool that supports heterogeneous behaviors and real-time interactivity. By exposing perception weights as runtime parameters, users can dynamically modulate the crowd's behavioral trade-off between physical collision avoidance and implicit environmental preferences. The framework also supports real-time editing of suitability maps, allowing designers to modify implicit environmental rules on the fly to simulate dynamic preference changes without retraining. This dual flexibility enables advanced emergent behaviors, such as reciprocal crowd-environment interactions, where agent movement progressively alters the suitability map (e.g., forming desire paths), creating a continuous feedback loop between the crowd and its surroundings. To confirm the practical efficacy of these capabilities, we include a user study to validate the system's ability to replicate real-world crowd behaviors through perception-guided authoring.

The main contributions of this paper are:

- **A Multi-Perception Fusion Architecture:** We propose a specialized neural network architecture that integrates Entity Perception for collision avoidance with Implicit Perception for semantic navigation, to balance both behaviors without collision avoidance overruling semantic navigation.
- **Modular Two-Phase Training Strategy:** We introduce a decoupled training pipeline combined with curriculum-based env-templates to resolve the optimization conflicts between hard physical constraints and soft environmental preferences, significantly improving generalization to unseen environments.
- **Interactive Authoring & Emergent Behavior:** We demonstrate a workflow that allows for real-time environmental editing (e.g., painting soft constraints) and facilitates emergent social phenomena, such as trace-based path evolution.

This paper is organized as follows: In Section 2, we review previous related work. Then we present the details of our novel

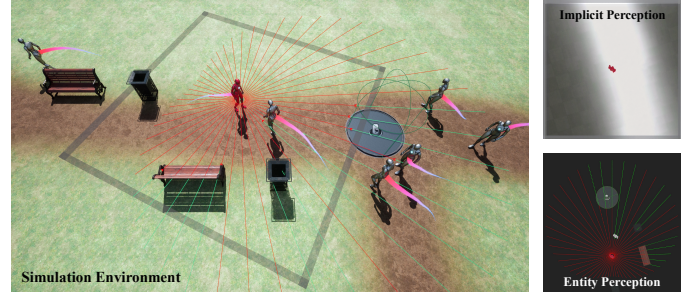


Fig. 1. Overview of our multi-perception decision-making framework. The left image shows the simulation environment with agents navigating obstacles using implicit and entity perception. The top-right image visualizes implicit perception, where the bitmap intensity represents movement preference. The bottom-right image illustrates the entity perception, using raycasting to detect surrounding objects.

method in Section 3, and the experimental results and analysis in Section 4. Finally, in Section 5, we discuss the implications of this work and outline future research directions. The source code, trained policy, and environments are available at: [link will be provided after the review process].

II. RELATED WORK

Significant progress has been made in crowd simulation research, encompassing macroscopic and microscopic approaches [20]. Macroscopic methods typically control the crowd as a whole, limiting their ability to manage diverse agent behaviors and complex environmental perceptions. Therefore, our focus is primarily on microscopic research. In recent years, deep learning based techniques have gained popularity in crowd simulation. Our review of related work emphasizes two key areas: coupling environmental perception with simulation, and reinforcement learning for crowd simulation.

A. Coupling Environment Perception with Simulation

In crowd simulation, one of the most fundamental perception tasks is detecting the static and dynamic entities to ensure local collision avoidance. In early approaches, agents perceived environmental elements through simple distance and relative position measurements, and made decisions based on predefined rules, like in the Boids model [4] and the social force model [7]. Additionally, in various velocity-based methods, agents perceive the positions and velocities of potential colliders to select movement speeds from a safe velocity domain [21], [22]. However, these methods rely on abstract mathematical states rather than sensory inputs, oversimplify the perception of the environment, limiting it to detecting agents and often other static obstacles such as walls.

To bridge the gap between abstract states and realistic human sensory processing, vision-based perception has become increasingly popular. For example, using optical flow from a simplified render of the environment to drive the computation of agents' movement to avoid collision [23]. Dutra et al. [24] proposed a cost function for evaluating the danger perceived from the synthetic vision component. A virtual camera can be used to extract optical flow and

depth images of the scene to perform complex navigation tasks for the agents [25]. Although vision-based methods offers rich environmental perception, it is often computationally intensive. As a balanced alternative between computational efficiency and sensory realism, raycasting is frequently used as a simplified alternative to simulate perception [26], [27]. Raycasting approaches typically launch a fan of uniformly distributed rays and compute the distance to colliders based on the intersection point between each ray and the closest collider.

In addition, perceiving the temporal context of crowd dynamics is crucial for generating reasonable behaviors. Data-driven methods enable agents to learn from real-world examples, thereby maintaining a natural behavior without explicit models. For example, agents perceive the trajectories of others (as a sequence of positions and velocities), which is subsequently used to query the next trajectory from the example database [11]. Charalambous et al. [28] encodes the situations perceived by individuals into Temporal Perception Pattern (TPP), which is a combination of a visibility patterns over several simulation frames. Then it designs a Perception-Action-Graph (PAG) to accelerate the search for example patterns during the simulation process. Yan et al. [29] simulated more natural pedestrians from real trajectories using a transformer-based GAN.

To support global motion planning, agents require a structured representation of the navigable space. Cellular automata (CA) models [30] segment the environment into a regular grid of cells, enabling agents to simulate perception by accessing the discretized environmental information. For continuous space guidance, flow fields can be used to drive the agents' movement by replacing their current goal with the new one introduced by the field of vectors [31], [32]. This approach enables fast computation of large crowds but has the limitation that the entire crowd's movement is simultaneously governed by the same vector field; consequently, supporting individual goals requires creating a separate field for each agent. To establish flexible local rules, researchers have proposed the implementation of smaller vector fields to strategically influence the local movement of the agents [33].

Besides the geometry-based [1] and cell-based [34] environmental modeling which define "where agents can go" (traversability), some researchers have also used grayscale bitmaps to model environmental information to encode "where agents prefer to go" (semantics). For example, Wang et al. [35] used human-activity maps to represent human activity intensity in architectural spaces to design 2D floor plans. Oshita [36] represented perceived agent positions using grayscale maps and then applied CNNs to extract global environmental features for motion decision-making. Jordão et al. [37] encoded crowd density information as grayscale maps, constructing virtual crowd distributions based on *Crowd Patch* [38]. McIlveen et al. proposed pedestrian environment designer (PED) [39], in which multi-layer grayscale maps were used to represent environmental rules such as entrances, obstacles, and attraction points, which were then integrated with vector-field-based methods to control crowd movement. These grayscale-based methods primarily represent explicit control rules. However, current research lacks a method for modeling **implicit rules**

that capture crowd movement preferences, serving as **soft constraints** to alter the agent's paths but without replacing the agent's goal in the environment.

B. Reinforcement Learning for crowd simulation

Reinforcement Learning (RL) provides a robust framework for agents to learn sequential decision-making strategies by optimizing cumulative rewards [40]. While value-based deep RL methods like Deep Q-Networks (DQN) [41] demonstrated the potential of RL, they are limited to discrete action spaces. For continuous crowd control, Proximal Policy Optimization (PPO) [42] has become a common default choice due to its balance between sample efficiency and training stability. Therefore, we adopt PPO to learn continuous navigation policies that seamlessly integrate collision avoidance with environmental preferences.

In the domain of crowd simulation, RL has been widely applied to collision avoidance and navigation tasks. For instance, Lee et al. [43] employed DQN with raycasting-based perception to train adaptable agents. Xu et al. [44] utilized Direct Agent Perception (AP), i.e., accessing the positions and velocities of neighbors, combined with supervised learning to enhance behavioral realism. While these methods effectively address collision avoidance, they often treat the environment as static obstacles, lacking the flexibility to model complex environmental preferences.

Furthermore, improving the robustness of RL-based navigation to unseen layouts and evolving scene conditions remains a critical challenge in crowd simulation. To this end, some approaches augment RL policies with structured spatial priors that provide global guidance, thereby reducing sensitivity to local observation noise and helping generalization across different environment configurations. For example, Li et al. [34] combine deep RL with anisotropic fields to inject global prior knowledge for navigation in complex environments, and Sun et al. [45] investigate deep RL for crowd navigation in unknown and dynamic settings, highlighting the adaptability of learned policies when the environment cannot be fully specified in advance. These works highlight the necessity of exposing agents to diverse environmental configurations during training to ensure the adaptability of the learned policies.

Beyond basic navigation, recent research has focused on generating diverse and heterogeneous behaviors. Hu et al. [46] employed a parameterized reinforcement learning method to generate diverse trajectory via a single shared policy network, with each agent moving according to its preferred speed. Panayiotou et al. [16] modified the reward function weights for different agents, providing them with unique and configurable personalities as configurable crowd profiles (CCP). Building on this work, Charalambous et al. [17] used state distributions from real trajectory data to develop a data-driven reward function for learning complex and nuanced heterogeneous navigation in real crowds. Inspired by these mechanisms, we dynamically adjust perception weight parameters during training to achieve a diversity of navigation styles responsive to various environmental factors.

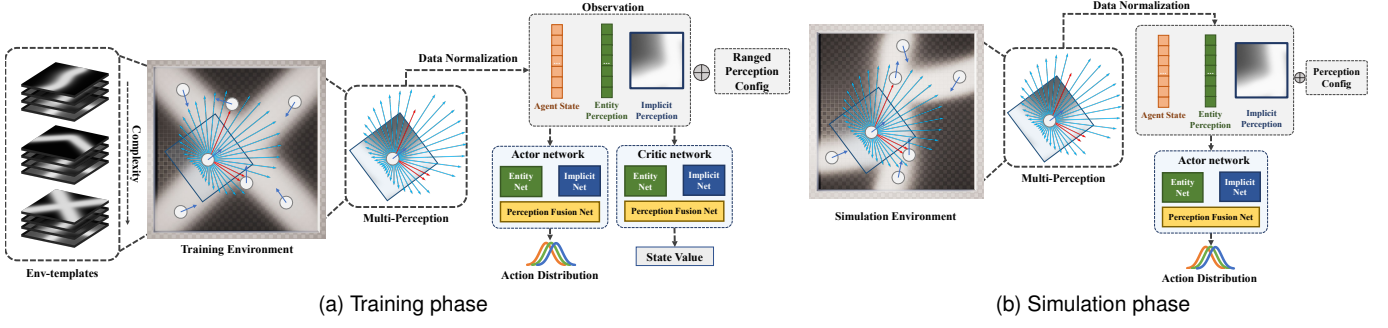


Fig. 2. **Overview of Our Proposed Framework.** The framework consists of a **training phase** and a **simulation phase**. During the training phase, Actor and Critic networks are trained using a **modular 2-phase training strategy**, incorporating curriculum-based environment templates and dynamically adjusted perception parameters. In the simulation phase, the agent's entity-implicit perception of the environment is fed into the Actor network as input, which then outputs action data to control agent movement.

III. METHODOLOGY

We propose a Multi-Perception Reinforcement Learning framework that explicitly decouples physical collision avoidance from semantic environmental preferences. This design enables agents to learn robust policies that balance hard and soft constraints without the labor-intensive and brittle parameter tuning required by traditional simulation methods. Formulating the crowd simulation task as a Markov Decision Process (MDP), we optimize the agent policy using standard Proximal Policy Optimization (PPO) [42] to maximize the expected cumulative return.

Unlike monolithic end-to-end approaches, we leverage a multi-perception state space processed via a dual-branch fusion network, which allows agents to dynamically weigh hard constraints against soft constraints. To ensure robust generalization across complex topologies, we employ a modular two-phase curriculum training strategy. This approach effectively mitigates the gradient dominance issues often observed in joint training, where one perception module suppresses the other, thereby preventing performance degradation. An overview of our system is shown in Figure 2.

The remainder of this section details the core components of our framework. Section III-A introduces the design of the Multi-Perception module. Section III-B defines the agent's State and Action spaces within the RL formulation. Section III-C elaborates on the Reward function design for balancing conflicting objectives. Finally, Section III-D presents the network architecture and our training strategy.

A. Multi-Perception Design

Our multi-perception approach combines an implicit perception module that detects soft environmental rules with an entity perception module to detect potential physical colliders (Figure 3a).

1) *Entity Perception*: To model the **entity perception**, we adopt the widely used raycasting approach. Each agent emits 50 evenly spaced rays in a 360° range centered on its current orientation. To simulate human behaviors such as gaze shifts [6] and attentional focus when perceiving the environment, we

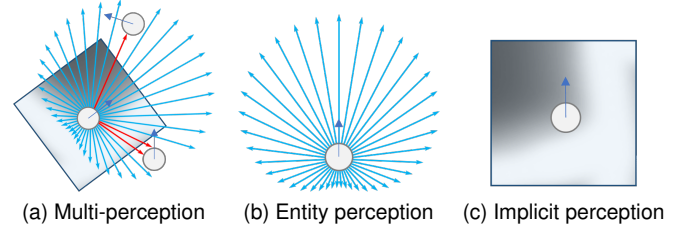


Fig. 3. Proposed multi-perception modalities.

calculate the maximum detection length for each ray using Equation 1.

$$l_i = \left[\lambda_a + (1 - \lambda_a) \frac{1 + \cos(\varphi_i)}{2} \right] \cdot l_{maxray}, \quad (1)$$

where l_i represents the maximum detection length of the i -th ray, φ_i is the angle the direction of the casting ray and the agent's gaze orientation, λ_a is a hyperparameter controlling the baseline attention level, with a default value of 0.2, and l_{maxray} denotes the maximum range of the rays, set to a default value of 10 m. Rays aligned with the agent's gaze direction, which simulate vision, are given greater attention and thus have a longer detection range (see Figure 3b). Conversely, rays in the direction directly opposite to the gaze have the shortest detection range, though nonzero. We use this shorter rays to simulate the fact that humans can sense objects behind them through sounds or intuition, but the impact of those entities on the forward trajectory of the agent is less relevant. To distinguish observed entities, similar to [45], we represent the raycasting results using the normalized detection distance and the relative velocity of the detected entities with respect to the agent. Please note that the term entity is not limited to other agents but also includes other obstacles that we may need to represent in the environment, such as walls or cars. The normalized detection distance is defined as the ratio of the detected distance d_i to the maximum detection range of the corresponding ray l_i , i.e., d_i/l_i , ensuring values are scaled between 0 and 1. These results collectively form the agent's perception of surrounding entities.

2) *implicit Perception*: To model **implicit perception** of environmental rules, we define a suitability function $s(\mathbf{p})$ that

assigns a suitability value between 0 and 1 to each point $\mathbf{p} \in \mathbb{R}^2$. A value of 0 indicates an area to be avoided, while values close to 1 represent highly suitable regions for movement. To provide agents with local perception of these suitability values, we introduce a suitability map $\mathbf{o}_{implicit}$, which is a discretized representation of $s(\mathbf{p})$. This map is sampled within a fixed window around the agent, centered on its current position and aligned with its gaze direction, and represented as a standardized 32×32 grid to ensure consistent data processing. The resulting suitability map serves as the agent's perception of the implicit environmental rules, as illustrated in Figure 3c.

B. Agent State and Actions

Agents are modeled as circular disks with a radius of 0.5 m, at position $\mathbf{p}$ with velocity $\mathbf{v}$. Agents want to move towards a goal position $\mathbf{p}_g$. Egocentric observations [15] are employed, which represent positions and angles relative to a local coordinate system centered on the agent's location and aligned towards its facing direction. Building on our multi-perception design, we define the agent state in reinforcement learning as:

$$S = (O_{implicit}, O_{entity}, O_{agent}, O_{weights}) \quad (2)$$

where $O_{implicit} \in \mathbb{R}^{32 \times 32}$ represents the implicit perception of environmental suitability, using a fixed-size 32×32 grid centered around the agent; $O_{entity} \in \mathbb{R}^{150}$ encodes the perception of the surrounding entities, represented as a 150-dimensional vector derived from 50 raycasts, where each ray provides the normalized detection distance and the relative velocity of the detected entity; $O_{agent} \in \mathbb{R}^5$ describes the agent's own state, including its velocity $\mathbf{v}_{rel} = \mathbf{R}_\phi \cdot \mathbf{v}$, orientation ϕ , and relative position to the goal $\mathbf{p}_{rel_goal} = \mathbf{R}_\phi(\mathbf{p}_g - \mathbf{p})$, $\mathbf{R}_\phi$ represents the rotation matrix aligned with the current orientation of the agent; and $O_{weights} \in \mathbb{R}^2$ contains the perception weights ($w_{implicit}, w_{entity}$), which balance the influence of implicit and entity-based perceptions in decision-making. This state, represented as a structured observation space, allows the agent to perceive and interact with its environment effectively, integrating both implicit environmental preferences and explicit entity information.

Unlike most existing methods that align the agent's orientation with its movement direction, we decouple the gaze direction from the movement direction in the action space. This allows the simulation of more complex movement behaviors observed in real crowds, such as moving in directions other than the gaze direction or retreating in high-density scenarios. Specifically, we use acceleration (a_x, a_y) and angular velocity (ω) as the agent's continuous action space, denoted as $A = (a_x, a_y, \omega)$, where $A \in \mathbb{R}^3$. The value ranges for the agent's state and action spaces are detailed in Table I.

C. Rewards

Reinforcement learning optimizes behavioral strategies by maximizing the expected cumulative reward, making the design of a proper reward function critical. Our reward function

TABLE I
VALUE RANGE OF AGENT'S STATES AND ACTIONS.

Parameter	Value Range	Description
$s(\mathbf{p})$	$[0, 1]$	Cell value of suitability map
v_x	$[-0.5, 0.5]$	Speed perpendicular to the agent's orientation
v_y	$[-0.5, 1.6]$	Speed aligning with agents' orientation
a	$[-2, 2]$	Acceleration perpendicular to or aligned with the agent's orientation
$\omega_{angular}$	$[-0.25\pi, 0.25\pi]$	Angular speed of agents' orientation

consists of three main components: base motion rewards, entity perception rewards, and implicit perception rewards:

$$R(t) = R_{base}(t) + w_{implicit} \cdot R_{implicit}(t) + w_{entity} \cdot R_{entity}(t) \quad (3)$$

where $w_{implicit}$ and w_{entity} are dynamic per-agent weight parameters that allow diverse navigation style of crowd.

The **base motion reward** (R_{base}) includes four terms. The living penalty term R_{live} , applied every timestep, encourages the agent to keep moving and reach the destination as quickly as possible. The trajectory smoothing reward R_{smooth} penalizes sudden changes in velocity and direction of the trajectory, improving its naturalness and smoothness. Finally, given the distance d_g from the agent to the goal, the term R_{ag} incentivizes it to approach the goal and R_{rg} rewards reaching it.

$$\begin{aligned} R_{base}(t) &= R_{live}(t) + R_{smooth}(t) + R_{rg}(t) + R_{ag}(t) \\ \begin{cases} R_{live}(t) &= k_{live} \\ R_{smooth}(t) &= k_{smooth} \cdot |v(t) - v(t-1)| \\ R_{ag}(t) &= k_{gt} \cdot (d_g(t-1) - d_g(t)) \\ R_{rg}(t) &= k_{goal} \end{cases} \end{aligned} \quad (4)$$

The **entity perception reward** (R_{entity}) includes two terms that penalize the agent in case of collisions with other agents and with static obstacles: R_{ac} and R_{oc} , respectively.

$$\begin{aligned} R_{entity}(t) &= R_{ac}(t) + R_{oc}(t) \\ \begin{cases} R_{ac}(t) &= k_{ac} \cdot (\exp(r_1 + r_2 - d) - 1) \\ R_{oc}(t) &= k_{oc} \cdot \exp(t_{ov}/dt) \end{cases} \end{aligned} \quad (5)$$

where r_1 and r_2 are the radii of the two interacting agents, d is the Euclidean distance between their centers, t_{ov} represents the duration of the agent's overlap with a static obstacle, and dt is the timestep size, with a default value of 0.04 seconds.

The **implicit perception reward** ($R_{implicit}$) rewards the agent for moving in areas with higher preferences and penalizes it for moving in less suitable regions. Given a user-defined suitability threshold k_{thresh} , the suitability map $s(\mathbf{p})$ and the position $\mathbf{p}(t)$ of the agent at time t , we define it as:

$$R_{implicit}(t) = k_{implicit} \cdot (k_{thresh} - s(\mathbf{p}(t))) \quad (6)$$

The default values of the parameters k_i in the reward function are shown in Table II.

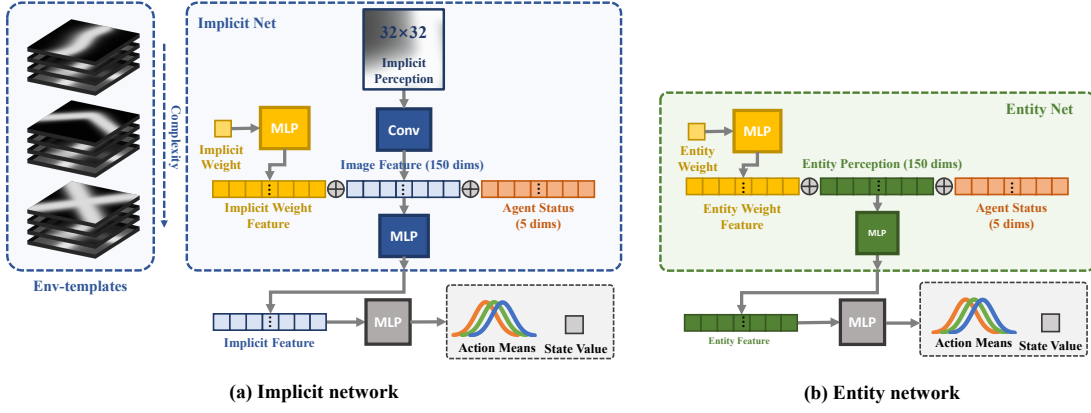


Fig. 4. Structures of the networks of independent module training in the first phase.

TABLE II
DEFAULT VALUE OF PARAMETERS IN OUR REWARD FUNCTION.

Parameter	Value	Description
k_{live}	-7.0	Living penalty
k_{smooth}	-0.25	Trajectory smoothness penalty
k_{ag}	100.0	Approaching goal reward
k_{goal}	1000.0	Reaching goal reward
$k_{implicit}$	-50.0	Implicit rule penalty
k_{ac}	-100.0	Agents collision penalty
k_{oc}	-1.0	Obstacle collision penalty
k_{thresh}	0.9	Penalty threshold of environmental rule

D. Network Structure and Training Strategy

We adopted a two-phase and modular training strategy. In the **first phase**, we separately train the entity perception and implicit rule perception modules, as shown in Figure 4. The input to the entity perception module consists of a 150-dimensional vector derived from the results of 50 raycasting detections and the agent's own state. The input to the implicit rule perception module is a 32×32 grid, processed through a multi-layer convolutional network for feature extraction, with the agent's own state also included as input to the network. Additionally, we design separate MLPs for the entity perception weight and implicit perception weight to enhance their influence within their respective network modules. We design nearly identical network architectures for the Actor and Critic networks. The only difference lies in their output dimensions: the Actor's output corresponds to the agent's action space, which is 3-dimensional, while the Critic's output consists of a single value representing the evaluation of the current state.

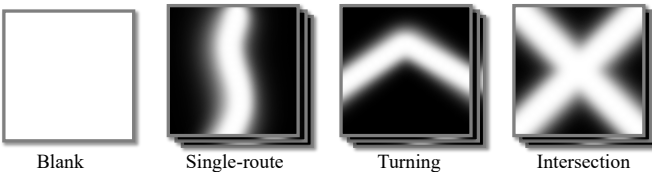


Fig. 5. Env-templates from simple to complex. Brighter areas represent higher suitability values.

During the training of the implicit perception module, we

employed a curriculum-based training strategy to ensure the model's adaptability to unknown and complex environments. We designed four sets of environmental templates with scenarios of increasing complexity, as shown in Figure 5. These sets correspond to blank, single-route (with varying curvatures), turning (with varying turning angles), and intersection (with varying rotation angles). The training process progresses from the simplest template set to the most complex. Each stage is initialized from the model trained in the previous stage and trained for 2000 episodes. The training objective of the implicit perception module is both goal-driven and focused on interactions with implicit environmental rules. Therefore, during training, we set the w_{entity} parameter to 0, eliminating the influence of the entity perception reward R_{entity} on the implicit perception module. For $w_{implicit}$, we initially sample random values from the maximum range $[0.8, 1.0]$, then from the minimum range $[0.0, 0.2]$, and finally adopt a curriculum-based assignment approach that gradually covers the entire range $[0.0, 1.0]$.

When training the entity perception module, we focus only on the entities in the environment. The primary objective is to train collision avoidance and goal-driven behaviors. Thus, training is conducted exclusively in the blank environment, with $w_{implicit}$ set to 0. The value assignment method for w_{entity} follows the same approach as $w_{implicit}$ during the implicit perception training, starting with random values in the maximum range, followed by the minimum range, and finally covering the entire range.

In the **second phase** of training, the goal is to enable agents to make motion decisions by integrating implicit environmental perception and entity perception. To achieve this, the two perception modules from the first phase are retained, and their model parameters are fixed. We additionally design a perception fusion module, which takes as input the outputs of the two pre-trained modules from the first phase. These inputs include the feature vector and corresponding weight features of the implicit perception module, as well as the entity perception module. Similar to the first phase, the Actor and Critic networks in the second phase share the same structure, except for the difference in output dimensions. The structure of the perception fusion module is shown in Figure 6.

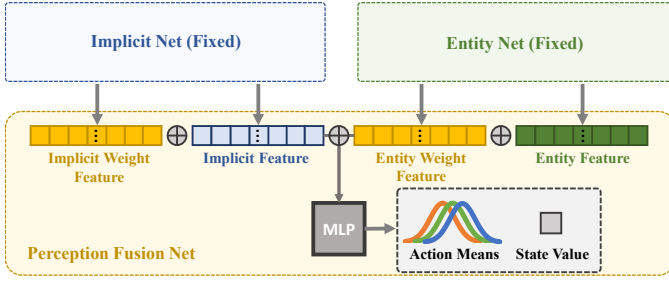


Fig. 6. Structures of the networks in perception fusion phase.

Since the weight parameters were already trained with a curriculum-based value strategy in the first phase and the weight feature networks are not updated in the second phase, we adopt a random assignment approach for both weight parameters across the full value range $[0.0, 1.0]$ during training. To prevent excessive dependence on specific templates, we empirically chose the **turning** as the training scenario, since it provided the best generalization. Subsequent experiments will show that this training strategy not only allows the model to adapt well to unknown environments but also enables the two perception weight parameters to effectively influence the agent's motion decisions.

For the basic training configuration, both phases adopt the same setup. The training environment is a square scene measuring $40\text{m} \times 40\text{m}$. At the beginning of each episode, 30 agents are randomly initialized within the scene and assigned target positions located at least 10m away from their starting positions, with implicit values greater than ϕ_{thresh} . An agent is considered to have completed its motion either when it reaches its destination or when it collides with a static obstacle for more than 2 seconds. This delay was empirically chosen as a balance between allowing time for agents to find a way to adjust their path and continue moving, and preventing agents from remaining stuck indefinitely. Immediate termination could cause premature dropout of agents still capable of making valid progress, especially in dense or complex environments.

Upon completion, the agent is replaced by a newly generated one at a random location in the scene. At the start of each episode, one agent is randomly selected as the target for monitoring; the episode ends when this agent completes its motion. All agents in the scene share the same training policy. Since PPO is an on-policy reinforcement learning model, the policy is updated every 256 timesteps or whenever an episode ends. Each update uses the Markov decision records of agents whose memory lengths reach the maximum of 256 as training data. A batch size of 64 and 8 epochs per update are used. After each update, all agent memories are cleared.

IV. EXPERIMENTS AND DISCUSSION

We used a laptop with an i9-12900HK CPU, an NVIDIA GeForce RTX 3080 Ti Laptop GPU, and 64GB of RAM as the experimental setup. We designed a system architecture that separates model training from simulation environment and visual rendering. The PPO model is implemented using Pytorch,

while multi-perception, reward calculation, and visualization rendering are implemented in Unreal Engine 5. The agent state and action data are transmitted between the two components via TCP. The hyperparameters used during training are shown in Table III.

TABLE III
DEFAULT VALUES FOR HYPERPARAMETERS.

Parameter	Value	Description
dt	0.04s	Simulation step
$\text{lr}(\text{actor})$	$2.5e-5$	Learning rate for actor network
$\text{lr}(\text{critic})$	$2.5e-4$	Learning rate for critic network
γ	0.99	Discount factor
epochs	8	Training epochs
batch size	32	Batch size
buffer size	256	Buffer size

A. Efficacy of Perception Modulation

To validate the effectiveness of w_{entity} and w_{implicit} , we used the Circle scenario in a blank template to test w_{entity} and a fixed Multi-route scenario to test w_{implicit} .

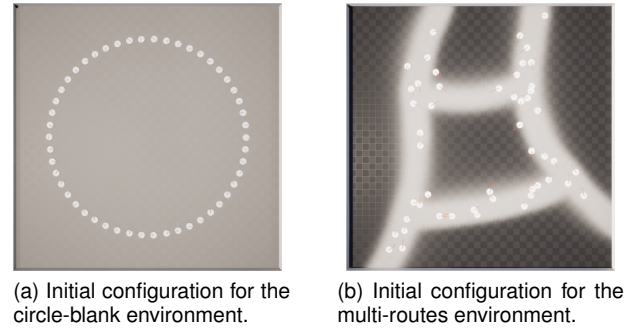
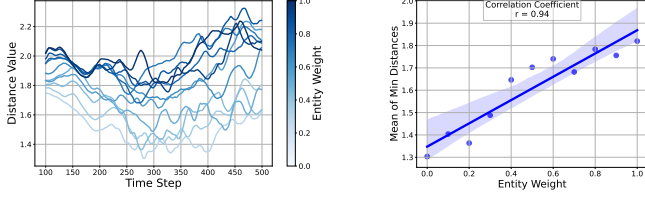


Fig. 7. Initial configurations for testing the perception weights.

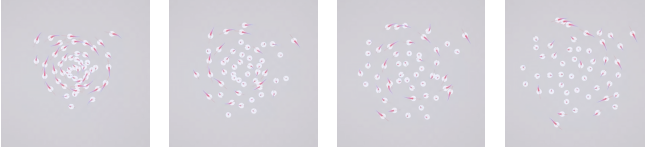
First, in the Circle scenario (Figure 7a), 50 agents are arranged in a circle with a radius of 15m , and each agent's target position is set to the opposite point on the circle. We fixed w_{implicit} to 1 and varied w_{entity} from 0.0 to 1.0 in increments of 0.1, conducting 11 rounds of experiments. In each round, we recorded the positions of all agents at every timestep and calculated the Distance to the Nearest Neighbor (DNN) for all agents. We plotted the average DNN of all agents over a fixed time range (as they approach the center of the circle) as a curve, see Figure 8a. The graph shows that the DNN values are lower for the lighter blue curves, which represent smaller w_{entity} values. Additionally, Figure 8b illustrates a linear correlation of 0.94 between the minimum distances observed during the highest-density phase and the w_{entity} value, indicating a strong positive correlation. This confirms the effect of w_{entity} on agents' entity perception: the larger w_{entity} , the stronger the entity perception, leading to better collision avoidance and larger DNN values. Conversely, smaller w_{entity} values result in weaker entity perception, leading to higher collision levels between agents and smaller DNN values.

Figure 9 shows a comparison of group movements at their highest density experienced in the circular scenario under different w_{entity} configurations.



(a) Average DNN in ranged w_{entity} over a fixed time range (crowd gather in center). (b) Minimum average DNN in ranged w_{entity} .

Fig. 8. Correlation between entity weight and the average DNN of crowds.



(a) $w_{\text{entity}} = 0.0$ (b) $w_{\text{entity}} = 0.3$ (c) $w_{\text{entity}} = 0.6$ (d) $w_{\text{entity}} = 1.0$

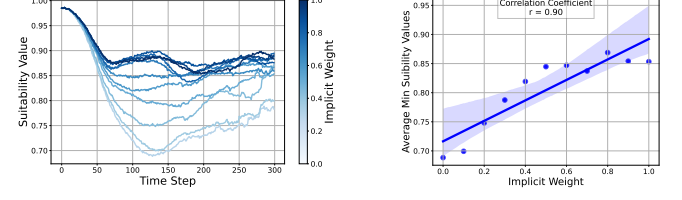
Fig. 9. Visualization of crowd behavior in the same initial circle environment (Figure 7a) under different w_{entity} values.

In the multi-routes scenario (Figure 7b), we followed a similar approach as above: setting w_{entity} to a fixed value of 1 and varying w_{implicit} from 0.0 to 1.0 in increments of 0.1, conducting 11 rounds of experiments. In each round, we recorded the suitability value of all agents at every timestep. We plotted the average suitability values over a fixed time range as a curve, as shown in Figure 10a. Lighter blue curves represent smaller w_{implicit} values, and the lighter the curve, the smaller the suitability values. Additionally, Figure 10b illustrates a linear correlation of 0.92 between the lowest average suitability value during each round and w_{implicit} , indicating a strong positive correlation. This validates the role of w_{implicit} in agents' implicit perception. The larger w_{implicit} , the stronger the influence of implicit perception, causing agents to exhibit a stronger preference for areas with higher suitability values. Conversely, lower w_{implicit} values result in agents paying less attention to implicit environmental perception, making them more likely to enter areas with lower suitability values. The comparison of group movements at the same moment in the Multi-routes scenario under different w_{implicit} configurations is shown in Figure 11.

Furthermore, our model allows dynamic adjustment of perception weights at runtime by modifying the corresponding input values, enabling real-time control over crowd navigation style. Additionally, each agent in the crowd can be assigned different perception weight configurations, thus enabling the simulation of diverse crowd behaviors. The specific effects of these adjustments are demonstrated in the supplementary video.

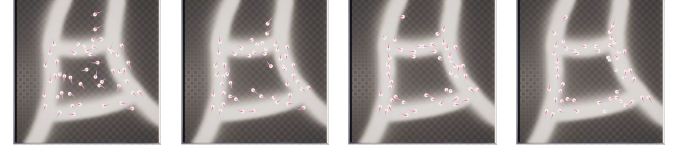
B. Analysis of Network Components

In this section, we conduct an ablation study to individually evaluate the impact on performance of each module in our proposed framework.



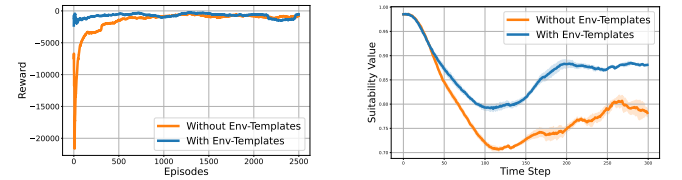
(a) Average suitability for different w_{implicit} . (b) Minimum average suitability for different w_{implicit} .

Fig. 10. Correlation between implicit weights and the average suitability of crowds.



(a) $w_{\text{implicit}} = 0.0$ (b) $w_{\text{implicit}} = 0.3$ (c) $w_{\text{implicit}} = 0.6$ (d) $w_{\text{implicit}} = 1.0$

Fig. 11. Visualization of crowd behavior in the same initial multi-routes environment (Figure 7b) under different w_{implicit} values. Brighter areas represent higher suitability values.



(a) Reward comparison

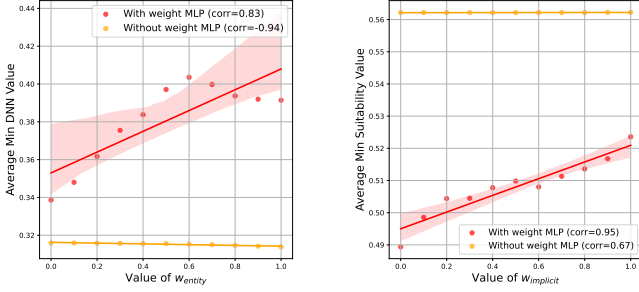
(b) Suitability value comparison

Fig. 12. Performance comparison of with/without curriculum-based Env-templates.

1) Env-Templates: We employ curriculum-based env-templates to improve the model's adaptability to unknown implicit environmental perceptions. For validation, we focus on the implicit perception module. Specifically, we compare the model trained progressively on four templates (blank, single-route, turning, and intersection, as shown in Figure 5) against a model trained directly on the intersection template. As shown in Figure 12a, the reward convergence scores of the two models are nearly identical, demonstrating comparable performance on the training set templates.

To further evaluate their performance, we tested the two models in the unseen Multi-routes environment (Figure 7b). Each model was run 10 times in the Multi-routes environment, and the minimum average suitability curve for each frame is plotted in Figure 12b, under the same environment and with w_{implicit} set to 1, the model trained with curriculum-based templates achieves higher average suitability value than the model trained directly on the crossing environment. This indicates that the curriculum-based environmental templates enhance the adaptability and generalization of the implicit perception module to new environments.

2) Weight MLP: To enable the perception weight parameters w_{implicit} and w_{entity} to influence the degree of implicit and entity perception and subsequently affect action decisions,



(a) Correlation between w_{entity} and DNN (b) Correlation between w_{implicit} and suitability value

Fig. 13. Comparison of weight correlations based on the use of weight MLP.

we incorporated a **Weight MLP** into the network design to enhance the impact of these weights. To validate the effectiveness of the Weight MLP, we compared the performance of models with and without it. The correlation between weights and evaluation metrics, including DNN and suitability values, was used as the evaluation criterion. In this experiment, we did not adopt the 2-phase training strategy. Instead, the entire network was trained simultaneously, with the two weight parameters randomly sampled across the full range $[0.0, 1.0]$ during training. After convergence, the two models were tested in the Circle and Multi-routes scenarios to evaluate the effects of w_{entity} and w_{implicit} , respectively. As shown in Figure 13, in both scenarios, the network with the Weight MLP exhibited a stronger positive correlation between the weights and the evaluation metrics. Additionally, the weights had a more significant impact on the evaluation metrics of the simulated crowd. This validates the effectiveness of the Weight MLP in embedding the perception weight parameters into the network.

3) *Modular 2-Phase Training*: To validate the impact of modular 2-phase training on model performance, we compared two models: one with the fixed entity perception and implicit environmental perception modules pre-trained during the first phase, followed by training only the perception fusion module in the second phase, and another model where all modules are trained simultaneously until convergence. The reward convergence of both models is shown in Figure 14. The model trained with modular 2-phase training achieves a higher converged reward than the model trained with all modules simultaneously.

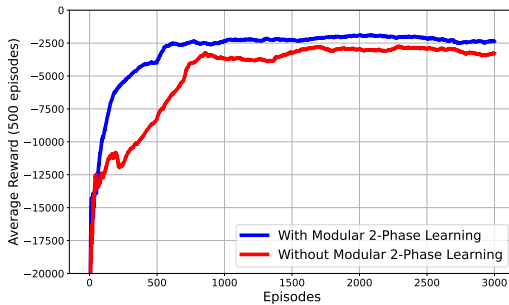
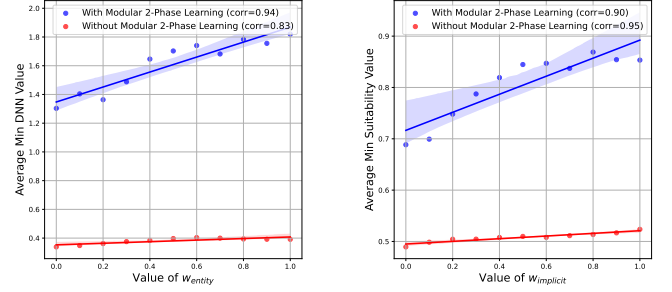


Fig. 14. Reward function comparison (W/Wo Modular 2-Phase Training).



(a) Correlation between w_{entity} and DNN (b) Correlation between w_{implicit} and suitability value

Fig. 15. Comparison of weight correlation based on the use of modular 2-phase training.

Additionally, we compared the effects of w_{entity} and w_{implicit} in the Circle and Multi-routes scenarios for both models. As shown in Figure 15, both models, equipped with the Weight MLP, exhibit high positive correlations between the perception weights and the behavioral evaluation metrics, regardless of whether modular 2-phase training was employed. However, the phased modular training model has a significantly greater influence on crowd behavior. A greater influence indicates that the perception weight parameters can more effectively adjust the agents' behaviors. Conversely, if the influence is small, agents with different weight configurations exhibit no noticeable behavioral differences during testing. From the comparison of regression curve slopes, the impact of w_{entity} on DNN is 6.78 times greater, and the impact of w_{implicit} on suitability values is 9.48 times greater. These results validate that modular 2-phase training not only enhances the model's performance but also amplifies the influence of perception parameters on agent behavior.

C. Adaptability and Generalization

In this section, we evaluate the adaptability and robustness of our proposed framework across three distinct dimensions of complexity: spatial generalization, crowd density, and heterogeneous social behaviors. We first demonstrate the model's ability to handle unseen environments without retraining. Next, we assess its collision avoidance performance under high-density stress. Finally, we showcase how assigning heterogeneous suitability maps can induce emergent social norms, such as bidirectional lane formation.

1) *Generalization to Unseen Environments*: To validate the adaptability and generalizability of the model, we designed a series of implicit environmental scenarios that are not included in the training set. With both w_{entity} and w_{implicit} set to 1, the corresponding crowd's behavior is shown in Figure 17. As observed in the unseen multi-routes environment, agents not only exhibit effective collision avoidance behavior but also move within areas with higher suitability values, demonstrating the model's adaptability.

Our framework distinguishes between navigation preferences (soft constraints), derived from implicit perception, and physical barriers (hard constraints), detected by the entity perception module. To validate the model's adaptability in

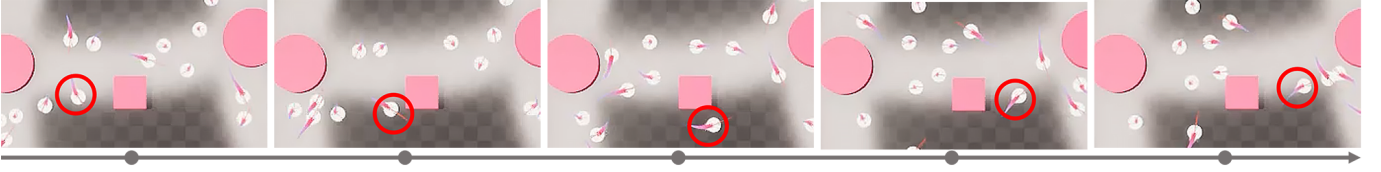


Fig. 16. Keyframes of an agent choose to move into low suitability area to avoid collision with static obstacle.

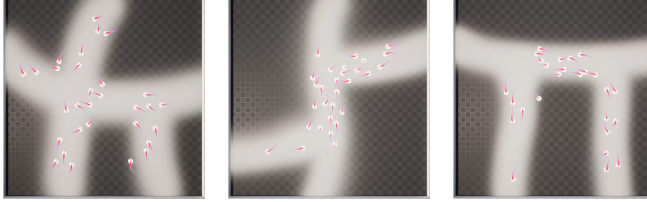


Fig. 17. Visualization of crowd behavior in unseen environments.

complex environments featuring hybrid constraints and dynamic agents, we designed specific test scenarios. As shown in Figure 16, agents successfully prioritize safety over preference, temporarily traversing lower-suitability areas to avoid collisions. This confirms the model’s robustness in resolving conflicting objectives between hard and soft constraints.

2) *Heterogeneous Crowds*: Our framework design allows for a wide range of configurations of weights and combinations of suitability maps. It can naturally support the simulation of heterogeneous crowd behaviors by assigning each agent a distinct configuration of weights or different sets of suitability maps, thereby enabling the representation of diverse navigation preferences. For example, a suitability map representing shaded areas can be included, with only a subset of agents assigning a high weight to this map, while others navigate without being influenced by it. Alternatively, different suitability maps can be used to model outdoor environments, such as grass, steep terrain, or puddles, and these maps can be selectively assigned to different agents. Several examples of such configurations and combinations are shown in the accompanying videos.

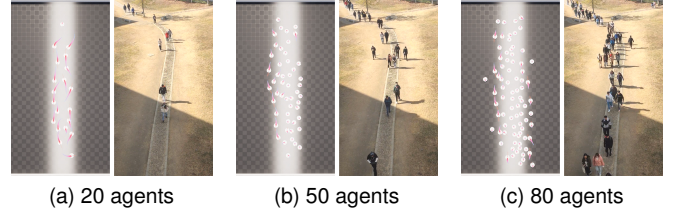
3) *Robustness in High-Density Crowds*: To assess robustness under extreme congestion, we simulated a bidirectional narrow path scenario with 80 agents. With full weight on both perception modules ($w_{\text{entity}} = 1, w_{\text{implicit}} = 1$, see Figure 18a), the crowd exhibits reduced speed and congestion. Agents prioritize high-suitability regions (the path) while maintaining strict collision avoidance, occasionally causing individuals to wait in lower-suitability areas when the central path becomes saturated. Reducing w_{implicit} to 0 (Figure 18b) encourages the crowd to utilize the entire available space, effectively bypassing the congested center. Conversely, setting w_{entity} to 0 (Figure 18c) diminishes collision avoidance behaviors, causing agents to force their way through the high-suitability zone despite collision risks. These comparisons demonstrate that adjusting perception weights effectively modulates the trade-off between navigation safety and environmental preferences.

Furthermore, we investigated adaptive behaviors across



(a) $w_{\text{entity}}=1, w_{\text{implicit}}=1$ (b) $w_{\text{entity}}=1, w_{\text{implicit}}=0$ (c) $w_{\text{entity}}=0, w_{\text{implicit}}=1$

Fig. 18. Comparison of crowd navigation preferences under different perception weights in a high-density bidirectional hallway.



(a) 20 agents (b) 50 agents (c) 80 agents

Fig. 19. Comparison of simulated and real crowd distributions under different densities in a bidirectional narrow path scenario (images from a publicly available video in bilibili).

varying crowd densities (20, 50, and 80 agents). We configured $w_{\text{entity}} = 1$ and a reduced $w_{\text{implicit}} = 0.25$ to encourage flexible utilization of lower-suitability regions to improve flow. Figure 19 compares our simulation results with real-world footage, revealing similar bidirectional flow patterns. Under low-density conditions (Figure 19a), agents predominantly occupy the high-suitability path. As density increases (Figures 19b and 19c), a *spillover effect* emerges: agents progressively spread into lower-suitability areas to resolve conflicts and maintain progress. These findings suggest that our model exhibits adaptable navigation preferences responsive to crowd density.

D. Interactive Authoring and Dynamic Scenarios

A practical crowd simulation framework must offer intuitive tools for artistic control and complex environmental modeling, going beyond mere navigation robustness. Our multi-perception architecture inherently supports this interactivity by utilizing suitability maps as dynamic, modifiable input channels. This design enables human-in-the-loop simulation, allowing users to edit, compose, or evolve environmental rules in real-time without altering the underlying agent policy. In this section, we demonstrate these capabilities through three distinct scenarios: real-time constraint authoring, compositional multi-preference navigation, and emergent reciprocal interactions between the crowd and the environment.

1) *Real-time Map Authoring*: We demonstrate the system’s capability for real-time constraint editing. By dynamically

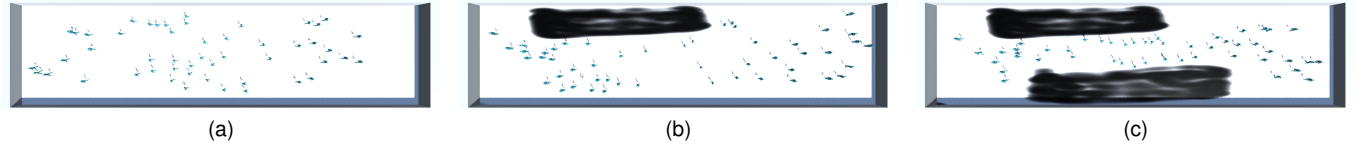


Fig. 20. Sequence of real-time map authoring in a hallway ($w_{\text{entity}} = 1, w_{\text{implicit}} = 1$). (a) Agents navigate freely across the wide corridor. (b) A user paints a soft-constraint to indicate a low-suitability area, and the crowd starts shifting the flow towards the lower passage. (c) A second soft-constraint is added, forcing the crowd to navigate towards the center without retraining.

modifying the suitability map O_{implicit} (painting with a digital brush), users can impose temporary restrictions or define new environmental rules on the fly. This capability is particularly valuable for simulating dynamic scenarios, such as sudden fire accidents or traffic control interventions. As illustrated in Figure 20, we simulate a hallway scenario with perception weights fixed at $w_{\text{entity}} = 1$ and $w_{\text{implicit}} = 1$. Initially, agents travel in opposite directions spread along a wide corridor (Fig. 20a). When the user paints a low-suitability area in the upper section (visualized in black in (Fig. 20b), agents moving in either direction immediately perceive this environmental change and shift their trajectories toward the lower passage to avoid it. Then, the user sketches a second constraint in the lower part, forcing the agents to adapt again to this evolving environment by navigating through the remaining high-suitability zones in the center (Fig. 20c). This experiment validates the system’s efficacy for real-time authoring, confirming that our agent policy generalizes effectively to dynamic spatiotemporal changes without requiring retraining. Please refer to the supplementary video for the dynamic painting demonstration.

2) *Multi-map Composition*: Our framework facilitates multi-preference navigation by decoupling environmental rules into independent semantic layers. Users can synthesize the final suitability input O_{implicit} via a linear combination of elementary maps, enabling agents to balance conflicting preferences without altering the underlying policy. Formally, the composite suitability value $S_{\text{final}}(p)$ at position p is computed as the weighted sum of N layers, constrained to the unit interval $[0, 1]$:

$$S_{\text{final}}(p) = \text{clamp} \left(\sum_{i=1}^N \omega_i \cdot S_i(p), 0, 1 \right) \quad (7)$$

where $S_i(p)$ represents the i -th preference layer and ω_i denotes its influence weight. We demonstrate this capability in a scenario combining three preferences: path following, shadow seeking, and puddle avoidance. As shown in Figure 21, the final map is synthesized as $S_{\text{final}} = \omega_{\text{path}} S_{\text{path}} + \omega_{\text{shadow}} S_{\text{shadow}} - \omega_{\text{puddle}} S_{\text{puddle}}$. In the baseline setting (Fig. 21a), agents strictly follow the path geometry. When the shadow preference is enabled (Fig. 21b), the crowd shifts towards the shaded outer edge while remaining on the path. Conversely, introducing the puddle layer (Fig. 21c) forces agents to navigate around hazardous areas. Finally, Figure 21d demonstrates the fusion of all three soft constraints: agents seek shade and follow the path while bypassing puddles. This confirms that adaptive, nuanced behaviors can be achieved through the simple algebraic composition of semantic features.

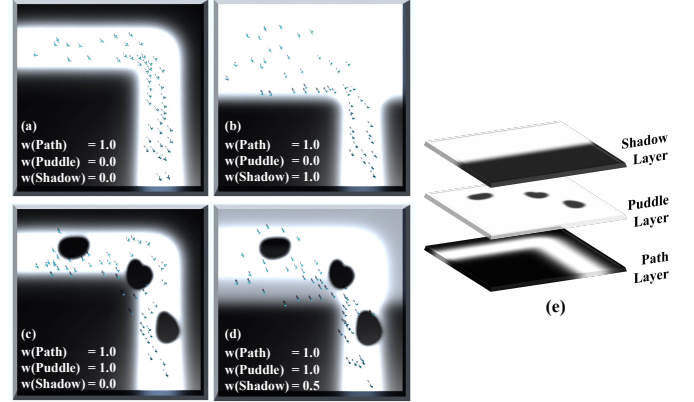


Fig. 21. Multi-preference navigation via map composition. (a) Baseline: Agents follow the path with $\omega_{\text{path}} = 1$. (b) Shadow Seeking: Agents prioritize the shaded outer curve ($\omega_{\text{shadow}} = 1$). (c) Hazard Avoidance: Agents bypass puddles ($\omega_{\text{puddle}} = 1$, acting as a penalty). (d) Composite Behavior: Agents balance path following, shadow seeking, and puddle avoidance simultaneously. (e) Visualization of the decoupled semantic layers: Shadow, Puddle, and Path.

3) *Reciprocal Crowd-Environment Interaction*: Finally, we demonstrate the system’s capacity for **emergent interaction**—a mechanism of indirect coordination where agents modify the environment, which in turn influences the future behavior of the group. Unlike traditional simulations where the environment is static, our framework supports a bidirectional relationship: the crowd not only perceives the suitability map but physically alters it through its movement.

We demonstrate this phenomenon on a Turning Path scenario featuring bidirectional goal-driven agents navigating between opposite ends of the curve. We implemented a dynamic update rule: whenever an agent traverses a low-suitability area the suitability value at that location increases by a small amount ($\Delta S = 0.01$). At the start of the simulation, agents largely follow the high-suitability curved path. However, since the environmental rule functions as a **soft constraint** rather than a rigid barrier, agents are not strictly confined to the preferred zone. Consequently, initial deviations into the inner corner occur due to the stochastic nature of the RL policy and local collision avoidance maneuvers arising from bidirectional interactions.

As shown in Figure 22, these stochastic deviations leave faint traces of increased suitability (visualized as white trails). This initiates a **positive feedback loop**: subsequent agents perceive these modified areas as more viable, increasing the probability of traversal, which further reinforces the path. At $T = 20s$ (Fig. 22a), only minor deviations are visible. By

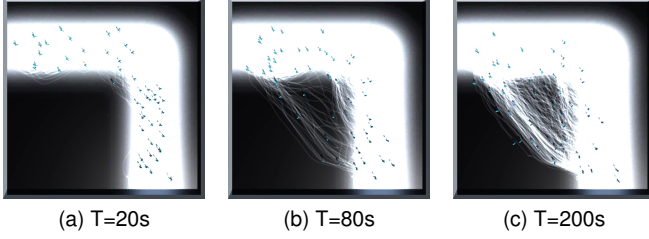


Fig. 22. Emergence of desire lines via stigmergy. The white trails represent accumulated suitability values ($\Delta S = 0.01$ per traversal). (a) Initial minor deviations. (b) A shortcut begins to emerge. (c) A fully self-organized trajectory is established.

$T = 80s$ (Fig. 22b), a shortcut begins to emerge. Finally, by $T = 200s$ (Fig. 22c), a distinct, self-organized trajectory is established, effectively replicating the organic evolution of “desire paths” observed in real-world landscapes such as parks. The complete evolutionary process is demonstrated in the supplementary video.

E. Perception Study

To evaluate the model’s capability in authoring crowd navigation styles, we conducted a perceptual comparison study using real-world crowd footage. We collected 20 video clips of real crowd scenes from publicly available sources and recreated each scenario using two different simulation settings: (1) **Crowd_{Basic-RL}**, in which agents were controlled using only collision avoidance behavior ($w_{\text{entity}} = 1, w_{\text{implicit}} = 0$), mimicking basic RL-based crowd simulation without implicit perception; and (2) **Crowd_{Multi-Perception}**, in which an expert user configured the weights for both entity and implicit perception. For each video, we reproduced the environment layout, agent spawn points, and goal locations in the simulation.

Anonymous participants, who consented to share their choices, were shown each real crowd video alongside the two corresponding simulations and asked to select which one more closely resembled the behavior observed in the real video. The interface of this perceptual test is shown in Figure 23.

The 20 video sets were presented in a randomized order to minimize bias. We had a total of 19 participants, thus 380 collected responses. From the responses gathered, we observed that, on average, 81.6% of participant selections favored **Crowd_{Multi-Perception}** as being more similar to the real crowd observed in the original video, while only 18.4% selected **Crowd_{Basic-RL}**. Figure 24 shows an aggregated summary of the participants’ selections across the different video sets; i.e., for each video set, we computed the percentage of times each crowd simulation setting was selected across all participants. Statistical analysis confirms this preference is highly significant: a paired t-test revealed a substantial difference in selection percentages ($t = -11.20, p < 0.001$), further supported by a Wilcoxon signed-rank test ($W = 1.00, p < 0.001$). These results indicate that configuring perception weights leads to more believable crowd behaviors, further validating the effectiveness of our model. The user study interface and responses are shown in the supplementary video.

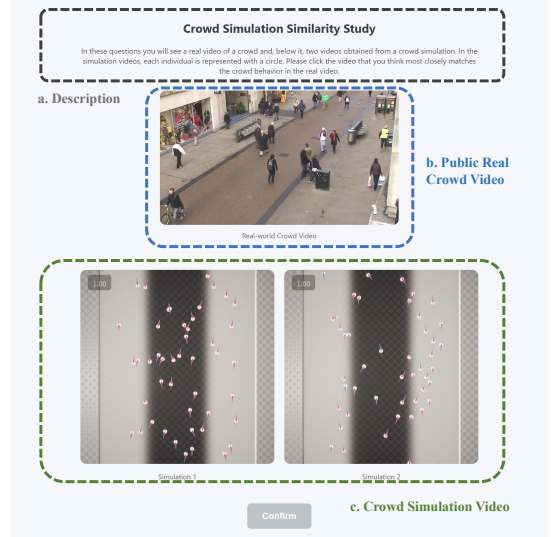


Fig. 23. Interface of the perceptual study. The interface consists of: (a) a description section explaining the task, (b) a real-world crowd video from public footage, and (c) two crowd simulation videos presented in random order. Participants are asked to select the simulation that most closely resembles the real crowd behavior.

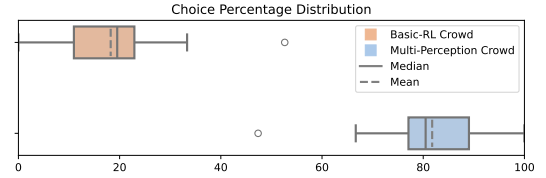


Fig. 24. Distribution of participant selection percentages for each crowd simulation across 20 video sets ($N = 19$ participants).

V. CONCLUSION AND FUTURE WORKS

This study proposes a novel reinforcement learning-based crowd simulation method that incorporates implicit environmental perception and entity perception to balance goal seeking and collision avoidance with environmental preferences. Suitability maps are used to simulate the preferences of real crowds for movement based on implicit environmental rules, such as personal habits and social norms. Simultaneously, raycasting is employed to detect the positions and velocities of surrounding entities for collision avoidance. After balancing the results of both perceptions, agents make the final motion decisions. We introduce a modular 2-phase training strategy. In the first phase, the entity perception and implicit environmental perception modules are trained separately. In the second phase, the parameters of these modules are fixed, and the perception fusion module is trained. During training, curriculum-based environmental templates are used to improve the adaptability of the implicit perception module to unknown and complex environments. Additionally, we designed two perception weight parameters and a corresponding Weight MLP structure. These weights dynamically cover their value ranges during training, enabling adjustments of the agent’s sensitivity to implicit and entity perception, thereby influencing its motion behavior. Experiments validate the strong positive correlation between changes in weight values and crowd motion eval-

uation metrics, such as Nearest Neighbor Distance (DNN) and Suitability values. Furthermore, ablation studies confirm the effectiveness of each training strategy. Moreover, tests in unseen environments outside the training set demonstrate the adaptability and generalization capabilities of the model. We have shown contributions that would be very difficult to replicate with existing approaches, showcasing the potential of our framework to simulate scenarios with dynamically changing environmental maps and weights, combining multiple maps, and even having reciprocal interactions between the crowd and the maps. Finally, we show a user study to demonstrate the capability of our system to serve as an authoring tool to better replicate observed real crowd movement.

The proposed model still has areas for further development. For instance, the hyperparameters in the current model are set empirically. In the future, optimization algorithms [47]–[49] could be employed to determine more appropriate values, thereby improving the model’s performance. Moreover, the model could be extended by matching agents’ perception parameters with their emotions and personality traits. Individuals with different personality traits could have distinct parameter configurations, enabling the simulation of more realistic and diverse crowd behaviors. These areas for improvement will serve as directions for our future research.

ACKNOWLEDGMENTS

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